# Problem statement

**Complete a partially observed bundle.**

$$\underbrace{S = \{\text{shirt, trousers}\}}_{\text{observed items}} \quad \longrightarrow \quad \underbrace{c \in \mathcal{I} \setminus S}_{\text{candidate completion}}$$

$$\boxed{\widehat{C}_K(S) = \mathrm{TopK}_{c \notin S}\ \theta(c \mid S)}$$

$\mathcal{I}$: item catalogue. $K$: number of recommendations.

**Goal:** rank items that belong with the observed combination.

# Problem statement

### Specific historical evidence

These items were selected together.

**Limitation:** some candidates have no recorded matches.

**Example:** a new shoe has no recorded co-occurrence with this shirt or these trousers.

### Transferable descriptive evidence

These items have related attributes.

**Limitation:** resemblance need not imply compatibility.

**Example:** trousers resembling the observed trousers may rank highly, even when the missing item is a shoe.

# Research question

**Can recorded bundle occurrences and item attributes support accurate completion without gradient-training a scoring model?**

1. **Use observed combinations**
   Distinguish several query items occurring together from separate pairwise matches.
2. **Handle missing item-pair observations**
   Score a candidate even when it has never occurred with the query items.
3. **Test the value of combining both**
   Compare with each source collection alone and with published neural results.

# Neural baselines

| Architecture | Parameters | Architecture summary |
|---|---|---|
| CLHE | 9.48M trained (POG) | Hierarchical attention with item- and bundle-level contrastive learning. |
| RaMen | 16.06M trained (POG) | Characteristic and graph encoders combined with a learned hypergraph. |
| Bundle-MLLM | 7B backbone | Multimodal item tokens projected into an LLM, adapted with LoRA. |

# BundleFuse

1. **Parameter-free scoring core**
   Zero gradient-trained scoring parameters.

2. **No neural-network training**
   Compute statistics and representations; select scalar settings on validation.

3. **Historical bundles as completion examples**
   Preserve joint evidence, transfer it through shared relations, and adjust reliance using candidate history.

# General scoring equations

**Final score:** mix all selected scores with a specified subset.

$$\theta(c \mid S) = \frac{f_c}{f_c + \kappa} Y_c + \frac{\kappa}{f_c + \kappa} s_0 P_c$$

| | |
|---|---|
| $Y_c$ | Combination of all selected score sources |
| $P_c$ | Category, descriptor or metadata scores, plus selected popularity; the chosen sources depend on the dataset |
| $f_c$ | Candidate's number of appearances in training bundles |
| $s_0, \kappa$ | Fallback scale and frequency threshold, respectively |

**At $f_c = 0$, only the scaled subset score $s_0 P_c$ remains.**

# General scoring equations

**Combine standardized sources $z_h$ with scalar weights.**

$$Y_c = \sum_{h \in \mathcal{H}} w_h z_h(c \mid S), \qquad P_c = \sum_{h \in \mathcal{P}} \widetilde{w}_h z_h(c \mid S), \quad \mathcal{P} \subseteq \mathcal{H}.$$

$\mathcal{H}$: available sources. $\mathcal{P}$: chosen fallback sources.

$z_h(c \mid S)$: standardized score from source $h$.

$w_h, \widetilde{w}_h$: scalar weights selected for the two combinations.

**The fallback combines selected sources already available to the full score.**

# General scoring equations

**Put different score sources on a comparable scale.**

$$z_h(c \mid S) = \frac{r_h(c \mid S) - \mu_h(S)}{\sigma_h(S) + \epsilon}$$

| | |
|---|---|
| $r_h$ | Raw score computed from history and available attributes |
| $\mu_h(S)$ | Mean of this score over catalogue items, for query $S$ |
| $\sigma_h(S)$ | Standard deviation over the same items |
| $\epsilon > 0$ | Small constant for numerical stability |

**No hidden completion labels enter these normalization statistics.**

# Results: POG

**Full-catalogue ranking on the supplied split**

| Method | Recall@20 | NDCG@20 |
|---|---|---|
| CLHE (published) | 0.02840 | 0.01930 |
| RaMen (published) | 0.03750 | 0.02260 |
| **BundleFuse** | **0.05446** | **0.03609** |

**+45.2% relative Recall@20 versus published RaMen**

# Results: Spotify

**Full-catalogue ranking on the supplied split**

| Method | Recall@20 | NDCG@20 |
|---|---|---|
| CLHE (published) | 0.10810 | 0.18060 |
| RaMen (published) | 0.10910 | 0.18820 |
| **BundleFuse** | **0.15565** | **0.25793** |

**+42.7% relative Recall@20 versus published RaMen**

# Results: why combine the evidence?

**NDCG@20 on the same queries within each dataset**

| Evidence | POG | POG_DENSE | Spotify |
|---|---|---|---|
| Historical only | 0.02094 | 0.25945 | 0.21170 |
| Shared only | 0.02929 | 0.07697 | 0.17191 |
| Combined, no fallback | 0.03394 | 0.27689 | 0.24320 |
| **Full BundleFuse** | **0.03609** | **0.28788** | **0.25793** |

**Neither source family alone matches the combined estimator.**

POG_DENSE full-model Recall@20: **0.39522**.